2.0.12

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Question

A product is made by mixing three raw materials 1, 2, 3 in varying proportions, where material 1 must account for not more than 50% of the total. If x, y and z are the amounts of raw materials 1, 2, 3, respectively, this constraint can be modeled as (2007 PI)

- $x \leq 0.5$
- $x \leq 0.5(x + y + z)$
- $0.5x \leq x + y + z$
- $x \geq 0.5(y + z)$

solution

Given,
Raw Materials: 1,2,3
Amount: x,y,z
let,
Total amount of Product: T

$$T = x + y + z \tag{1}$$

given constraint: Material one (x) must not account for more than 50% of the total.

$$x \leq 0.5(T) \tag{2}$$
$$x \leq 0.5(x + y + z) \tag{3}$$
$$x \leq y + z \tag{4}$$

Correct Option: b

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

y = np.linspace(0, 10, 20)
z = np.linspace(0, 10, 20)
Y, Z = np.meshgrid(y, z)

# The boundary condition: x = y + z (which is x = 0.5(x + y + z))
X_boundary = Y + Z

# Setup the figure
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')
```

Python Code

```
# 1. Plot the Boundary Plane
surf = ax.plot_surface(X_boundary, Y, Z, alpha=0.5, cmap='viridis
    ', edgecolor='none')

# 2. Visualize the Feasible Region (Constraint)
# We generate random points and keep only those where x <= y + z
x_pts = np.random.uniform(0, 20, 5000)
y_pts = np.random.uniform(0, 10, 5000)
z_pts = np.random.uniform(0, 10, 5000)

# Filter points that satisfy the constraint: x <= 0.5(x + y + z)
mask = x_pts <= (y_pts + z_pts)
x = np.linspace(0, 20, 30)
y = np.linspace(0, 10, 30)
z = np.linspace(0, 10, 30)
```

Python Code

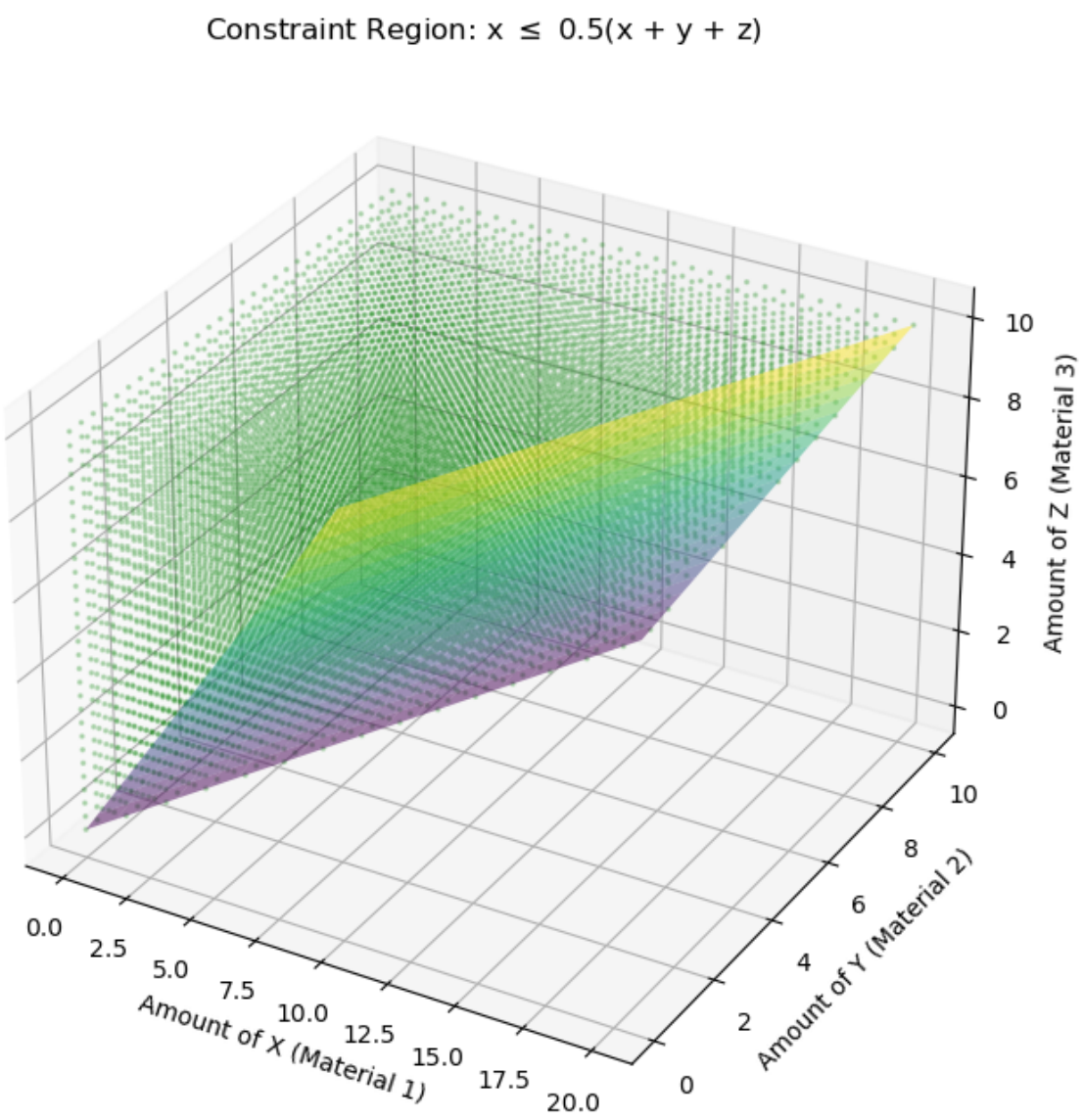
```
X, Y, Z = np.meshgrid(x, y, z)

mask = X <= (Y + Z)
ax.scatter(X[mask], Y[mask], Z[mask],
           color='green', s=2, alpha=0.2)

# Labels and Title
ax.set_xlabel('Amount of X (Material 1)')
ax.set_ylabel('Amount of Y (Material 2)')
ax.set_zlabel('Amount of Z (Material 3)')
ax.set_title('Constraint Region: x $\leq$ 0.5(x + y + z)')

plt.show()
```

Plot


Constraint Region: x ≤ 0.5(x + y + z)
Amount of X (Material 1)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Amount of Y (Material 2)
0
2
4
6
8
10
Amount of Z (Material 3)
0
2
4
6
8
10